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UNIT 1

📝 UNIT 1 PRACTICE ASSESSMENT: INFORMATION TECHNOLOGY AND SOCIETY

Programming & Artificial Intelligence

Egyptian Baccalaureate

Instructor: Eng/ Abanoub Refat

📝 UNIT 1 PRACTICE ASSESSMENT: INFORMATION TECHNOLOGY AND SOCIETY

Programming and Artificial Intelligence — Egyptian Baccalaureate (2nd Year)

First Semester — Duration: 3 Hours — Total Marks: 100

⚠️ STUDENT INSTRUCTIONS

- This assessment consists of three sections:
  - **Section A:** 50 Multiple-Choice Questions (50 Marks).
  - **Section B:** 3 Scenario-Based Application Questions (30 Marks).
  - **Section C:** 2 Conceptual Short-Answer Questions (20 Marks).
- All questions are based strictly on Pages 4–32 of the official **Programming and Artificial Intelligence (First Semester)** textbook.
- Answer all questions clearly. Show your analytical reasoning in Sections B and C.

SECTION A — MULTIPLE-CHOICE QUESTIONS (50 Marks)

Choose the single best answer for each question.

1. Which device, relying on power-hungry vacuum tubes, marks the "Birth of the Computer" stage (1940s–1960s)?

- A) Apple M1 Ultra
- B) ENIAC
- C) Intel 4004
- D) Pentium 4

2. What was the primary societal impact of computers during the first stage of IT history (1940s–1960s)?

- A) Personal computers became common in ordinary homes.
- B) Mobile Internet spread explosively worldwide.
- C) Computers were used almost exclusively by governments, militaries, and scientific institutions for heavy calculations.
- D) E-commerce and cashless payments became everyday routines.

3. Which technological leap allowed computers to shrink drastically in size and led to the personal computer (PC) boom in the 1970s–1980s?

- A) The commercialization of the Internet
- B) The invention of the microprocessor
- C) The transition from classical bits to qubits
- D) The development of centralized cloud datacenters

4. According to the textbook, during which decade did the commercialization of the Internet and the World Wide Web occur, globalizing information and spreading email?

- A) 1970s
- B) 1980s
- C) 1990s
- D) 2010s onward

5. What characterized the fourth stage of information technology (2000s)?

- A) The birth of room-sized machines like ENIAC
- B) The explosive spread of mobile Internet via smartphones
- C) The rise of large-scale cloud-based artificial intelligence
- D) The creation of the first office-desk personal computers

6. "IT delivered as a service over the Internet, backed by physical server machinery located in real datacenters" defines which concept?

- A) Edge Computing
- B) Cashless Payment
- C) Cloud Computing
- D) Quantum Computing

7. What is Moore's Law?

- A) The rule that a delay of 0.1 seconds in autonomous driving causes an accident.
- B) The principle that personal data collection must be authorized by users.
- C) The empirical observation that the number of transistors on an integrated circuit doubles approximately every two years.
- D) The scientific law stating that classical bits cannot hold more than one state at a time.

8. Why is transistor miniaturization in modern microprocessors approaching a physical limit?

- A) Because software applications are becoming too simple.
- B) Because of quantum tunneling and leakage current.
- C) Because physical server racks are running out of space in datacenters.
- D) Because users prefer cashless payments over credit cards.

9. What is the "quantum tunneling effect" in nanoscale silicon microprocessors?

- A) The process of routing internet data through subterranean physical cables.
- B) A subatomic phenomenon where electrons spontaneously slip through silicon barriers, disrupting chip reliability.
- C) The method by which a quantum computer processes classical bits in parallel.
- D) The transmission of data from an on-board edge device to a centralized cloud.

10. In chip manufacturing, what is "leakage current"?

- A) Unintentional electrical current escaping from chip circuits, causing excessive power consumption and heat.
- B) The transfer of private user coordinates to third-party advertisers.
- C) The slow backup of data from edge devices to the physical cloud.
- D) A surge in computing power caused by doubling transistors.

11. Which emerging technology utilizes qubits and the principle of superposition to speed up calculations that are difficult for traditional computers?

- A) Cloud Computing
- B) Virtual Reality (VR)
- C) Edge Computing
- D) Quantum Computing

12. Unlike a classical bit that holds a state of 0 OR 1, what does a quantum qubit hold?

- A) Only a state of 0, but with higher physical voltage.
- B) A combination of 0 and 1 simultaneously through the principle of superposition.
- C) Multiple decimal values ranging from 0 to 100.
- D) A sequence of pre-programmed hand-written rules.

13. Which of the following is the correct distinction between Augmented Reality (AR) and Virtual Reality (VR)?

- A) AR replaces the real world entirely; VR overlays digital details onto physical objects.
- B) AR overlays digital information onto real-world views; VR completely replaces the real world with an immersive virtual space.
- C) AR requires quantum superposition to run; VR runs strictly on physical cloud servers.
- D) AR is designed for autonomous driving safety; VR is designed for cashless mobile payment apps.

14. An autonomous car must make driving decisions instantly on-board without waiting for cloud communication because a delay of even _____ can cause an accident.

- A) 0.1 seconds
- B) 5.0 seconds
- C) 10.0 seconds
- D) 1.0 minute

15. Processing data instantly on a local device itself instead of sending it over the network to the cloud is known as:

- A) Parallel Quantum Processing
- B) E-commerce Scaling
- C) Edge Computing
- D) Cloud Virtualization

16. In Egypt, what represents the physical reality of "the cloud" in daily life?

- A) A series of invisible satellite networks floating in the atmosphere.
- B) Heavy physical server racks housed in real, high-security datacenters.
- C) On-board edge chips installed inside self-driving delivery vehicles.
- D) Virtual reality headsets worn by students during online learning.

17. A student checks an SNS app, pays for breakfast with a cashless app, joins a class online, and orders a book from an e-commerce shop. These activities represent:

- A) Technical limits of silicon microprocessors.
- B) The five social changes resulting from information technology.
- C) Pre-programmed steps in a conventional database.
- D) Examples of quantum mechanical superposition.

18. Which technology is defined in the textbook as a service allowing users to connect, post, and share information rapidly?

- A) E-commerce (EC)
- B) Social Networking Service (SNS)
- C) Cloud Datacenter Service
- D) Remote Work Platform

19. If a society realizes a fully cashless payment system, what is a major concern for a person who does not own a smartphone or bank card?

- A) They will experience a 0.1-second latency delay when purchasing food.
- B) They may face social exclusion and be unable to buy basic goods or services.
- C) Their personal files will be automatically uploaded to physical cloud datacenters.
- D) Their devices will suffer from leakage current and overheat.

20. What is Artificial Intelligence (AI) as defined generally in the textbook?

- A) A system that uses qubits to calculate scientific formulas at high speeds.
- B) A general term for technologies that reproduce or perform intelligent human behavior on a computer.
- C) A set of rigid hand-written rules programmed by software engineers.
- D) A technology that processes data strictly on a physical device without the Internet.

21. Which nested category hierarchy is technically correct according to the textbook?

- A) Machine Learning > Deep Learning > Artificial Intelligence > Generative AI
- B) Artificial Intelligence > Machine Learning > Deep Learning > Generative AI
- C) Deep Learning > Generative AI > Machine Learning > Artificial Intelligence
- D) Artificial Intelligence > Deep Learning > Generative AI > Machine Learning

22. What characterizes "Narrow AI" (Specialized AI)?

- A) It has general intelligence similar to a human and can perform any moral reasoning task.
- B) It is a mathematical model that operates without requiring any input data.
- C) It is highly expert at performing one specific task only, and does not possess general intelligence.
- D) It is an advanced quantum system that completely replaces physical cloud datacenters.

23. What is the fundamental difference between Conventional Programming and Machine Learning?

- A) Conventional programming uses quantum qubits; machine learning uses classical silicon bits.
- B) In conventional programming, the computer learns rules from examples; in machine learning, humans hand-write every rule.
- C) In conventional programming, humans write the rules; in machine learning, the computer automatically discovers rules and patterns from data and examples.
- D) Conventional programming requires massive cloud datacenters; machine learning runs strictly on edge devices.

24. "A computational system modeled after the workings of the nerve cells (neurons) of the human brain, which connects many components in layers to learn from data" defines a:

- A) Microprocessor Circuit
- B) Quantum Superposition Sphere
- C) Recommendation Engine
- D) Neural Network

25. Machine Learning is positioned as:

- A) A parallel technology to AI that does not require input data.
- B) One of the learning technologies that makes AI work by learning patterns from data.
- C) An advanced hardware chip designed to stop quantum tunneling.
- D) A system that generates text paragraphs using hand-written rules.

26. Which of the following is an example of Generative AI?

- A) A spam filter that automatically flags incoming emails as "junk."
- B) A smartphone camera that unlocks the screen by recognizing your face coordinates.
- C) A text-to-image software tool that generates a new digital painting from a written description.
- D) An automated routing algorithm that calculates the shortest delivery path for a truck.

27. What is "hallucination" in the context of Generative AI?

- A) A hardware error where silicon circuits leak electrical current.
- B) The phenomenon of AI generating content that sounds highly plausible but is factually incorrect.
- C) A psychological reaction experienced by users wearing virtual reality headsets.
- D) The mathematical process of averaging neural network weights to make a final judgment.

28. Why does a Generative AI tool (like a Large Language Model) create "hallucinations"?

- A) Because it is programmed with rigid conditional rules (IF/THEN statements).
- B) Because it predicts words based on statistical probability patterns rather than verifying factual records.
- C) Because it runs on edge devices that suffer from network latency.
- D) Because the model's training data has been completely erased.

29. Which everyday service is designed to predict user preferences from past behavior data and display personalized suggestions?

- A) Machine Translation
- B) Recommendation System
- C) Face Recognition
- D) Voice Assistant

30. Siri and Google Assistant are examples of which daily-life AI service?

- A) E-commerce (EC)
- B) Augmented Reality (AR)
- C) Voice Assistant
- D) Machine Translation

31. Which pair of applications are explicitly named in the textbook as examples of Machine Translation?

- A) Amazon and eBay
- B) Google Translate and DeepL
- C) Siri and Google Assistant
- D) ChatGPT and Midjourney

32. In what way does an "Image-diagnosis AI" assist medical professionals in healthcare?

- A) It automatically writes prescription rules without human doctor approval.
- B) It analyzes medical images like X-rays and CT scans to assist in detecting diseases.
- C) It optimizes the delivery routes of ambulances to eliminate traffic delays.
- D) It monitors soil moisture and predicts crop harvest timing.

33. A factory installs sensors to analyze machine data and predict hardware failures before they occur. This technology is called:

- A) Image Diagnosis
- B) Algorithmic Routing
- C) Predictive Maintenance
- D) Cashless Processing

34. How does AI assist the agricultural sector in Egypt?

- A) By automatically translating local dialects into classical Arabic.
- B) By analyzing field images to detect crop diseases and pests.
- C) By replacing physical tractors with quantum-powered drones.
- D) By processing credit card details at the local cooperative.

35. What is the "Black-box Problem" in modern AI applications?

- A) The physical destruction of database servers during power outages.
- B) A situation where it is unclear to humans how a complex AI model processed its input to reach a specific judgment.
- C) The high financial cost of importing Japanese microprocessors.
- D) A network error where data packets are trapped between edge and cloud systems.

36. Why does the "black-box" nature of deep learning models pose a risk in high-stakes fields like healthcare?

- A) Because edge devices do not have enough battery power to run them.
- B) Because doctors cannot verify the logical reasoning behind an AI's diagnosis to ensure it is safe and unbiased.
- C) Because they make computers run significantly slower than ENIAC.
- D) Because they cause agricultural drones to crash into crops.

37. While AI excels at finding patterns, what does it lack, necessitating human oversight for final decisions?

- A) Large-scale datasets and internet connections
- B) Microprocessors and parallel processing cores
- C) An understanding of ethics, responsibility, fairness, and privacy
- D) High-speed cashless payment APIs

38. "Bias in AI judgments caused by imbalances in the training data or system design" is called:

- A) Explanatory Error
- B) Algorithmic Bias
- C) Data Latency
- D) Superposition Deviation

39. Which of the following is a primary cause of algorithmic bias?

- A) The physical leakage current in the computer's CPU cores.
- B) The use of quantum qubits instead of classical bits.
- C) Training data that underrepresents certain demographics or reflects past human prejudices.
- D) Processing data at the local edge rather than in a centralized cloud.

40. In AI system design, what is a "proxy variable"?

- A) A backup processor that runs when the main computer fails.
- B) A variable that indirectly stands in for a protected attribute (such as using postal codes as a substitute for income or race).
- C) A temporary data file saved in a local scratch directory.
- D) A mathematical weight used to speed up edge computing latency.

41. How can a hiring AI be biased against female applicants even if "gender" is completely deleted from the dataset?

- A) By experiencing subatomic quantum tunneling.
- B) By learning from proxy variables (such as historical hobbies or school names) that correlate with gender.
- C) By running on centralized cloud physical datacenters.
- D) By generating plausible-sounding hallucinations.

42. Biometric facial recognition technology is considered "dual-use" because:

- A) It can run on both classical bits and quantum qubits.
- B) It improves public safety and convenience but also raises severe privacy concerns regarding mass surveillance.
- C) It operates in both agriculture and logistics with identical data inputs.
- D) It is used by both customers and small shop owners in cashless payments.

43. Which of the following describes "Explainable AI (XAI)"?

- A) An AI that explains to users how to purchase goods online using cashless payment apps.
- B) A technology that makes it possible for humans to understand why an AI model made a particular judgment.
- C) A system programmed with simple hand-written rules to write search-engine summaries.
- D) A programming model that teaches children how to write code.

44. Why is Explainable AI (XAI) considered an important ethical development?

- A) It allows developers to reduce the cost of physical cloud datacenters.
- B) It helps resolve the "black-box problem" by making automated decisions transparent and open to human audit.
- C) It automatically eliminates 0.1-second latency in autonomous vehicles.
- D) It replaces the need for human doctors in analyzing medical scans.

45. Which ethical principle of AI requires that the system does not unjustly discriminate against any particular person or group?

- A) Accountability
- B) Transparency
- C) Privacy Protection
- D) Fairness

46. Showing the AI's decision-making process, training data attributes, and inner workings clearly refers to which ethical pillar?

- A) Privacy Protection
- B) Transparency
- C) Accountability
- D) Fairness

47. What does the ethical pillar of "Accountability" require when an AI system makes an incorrect or harmful judgment?

- A) That the AI system automatically deletes its own database.
- B) That clear standards are established to identify who is legally and ethically answerable for the system's decisions and outcomes.
- C) That the system is migrated from edge computing to cloud computing.
- D) That the developer updates the training data with proxy variables.

48. Who is legally or ethically responsible when an AI makes an unfair or harmful decision?

- A) The AI software itself, as it acts autonomously.
- B) The classical bits that failed to process the mathematical weights.
- C) There is currently no universally established standard, making liability a complex debate among developers, operators, and users.
- D) The edge device, because it processed the data locally.

49. According to the textbook, if a school decides to use face-recognition cameras to record student attendance, what must they balance?

- A) Transistor counts under Moore's law versus quantum tunneling.
- B) Core clock speed versus parallel processing capabilities.
- C) Practical benefits (speed, convenience) versus student privacy concerns.
- D) Machine translation rules versus generative AI prompts.

50. The proper handling, security, and usage of personal data, giving individuals control over their information, is known as:

- A) Fairness
- B) Privacy Protection
- C) Explainable AI
- D) Remote Work Authorization

SECTION B — SCENARIO-BASED APPLICATION QUESTIONS (30 Marks)

Read each scenario carefully and answer the questions that follow, applying the concepts studied in Unit 1. Show your step-by-step reasoning.

> 📝 **Important Examination Note:** No external knowledge of Cairo’s logistics, Egypt's banking regulations, or Alexandria's school administrative bylaws is required or graded. Full marks are awarded strictly for applying Unit 1 concepts (such as latency/edge processing, algorithmic bias/proxy variables, and privacy/transparency/accountability) to the explicit facts provided in each scenario.

🗺️ Scenario 1: The Cairo Intelligent Express (Edge vs. Cloud Latency)

A logistics company in Cairo is launching a fleet of autonomous electric delivery drones. These drones fly at high speeds through congested urban streets to deliver packages. Each drone is equipped with high-resolution cameras that scan the surrounding airspace for obstacles (such as trees, power lines, other drones, and birds) to adjust its flight path in real-time.

- **Part A (10 Marks):** The engineering team must decide between two architectures:
  - **Option 1 (Centralized Cloud):** The drone streams live video feeds to a centralized physical datacenter, which calculates the avoidance coordinates and sends them back to the drone.
  - **Option 2 (Local Edge):** The drone processes the camera data locally on its on-board processors to calculate avoidance maneuvers.
  - *Task:* Using the concept of **latency**, the **0.1-second threshold in autonomous systems**, and the physical differences between **Cloud and Edge Computing**, evaluate both options and recommend the correct architecture. [15, multimodal_8]

⚖️ Scenario 2: The Egyptian Bank Recruitment Filter (Bias & Proxy Variables)

An Egyptian commercial bank receives over 50,000 resumes from university graduates every year for its entry-level training program. To save time, the bank's HR department purchases an AI-driven recruitment screening tool. The software is trained on the resumes of employees who were successfully hired by the bank over the past 20 years.

To ensure the system is completely "fair and non-discriminatory," the development team manually deletes the "Gender" column from the database before training. However, after three months of operation, the bank discovers that the AI systematically flags female applicants as "unsuitable" and filters them out.

- **Part A (10 Marks):** Explain how the AI was able to discriminate against female applicants even though the "Gender" column was completely deleted from the training data. Refer explicitly to the concepts of **algorithmic bias**, **unrepresentative/historical training data**, and **proxy variables**.

🛡️ Scenario 3: The Alexandria Smart Attendance Cameras (Privacy, Transparency & Accountability)

Alexandria Secondary School installs AI-powered face-recognition cameras at the school entrance gates. The system automatically scans the face of every student entering the campus, matches it against the registration database, records their attendance, and instantly sends an automated SMS alert to their parents' smartphones (e.g., *"Your child arrived at school at 07:45 AM"*).

One morning, the camera misidentifies a student, confusion occurs in the database, and the system records the student as absent. The student's parents receive an automatic alert that their child is missing. By the time the mistake is realized, the student has been issued an automated detention notice for skipping school. The school administration claims they are not to blame because "the computer made the decision on its own."

- **Part A (10 Marks):** Analyze this scenario using the four basic pillars of AI Ethics. Specifically:
    1. Identify which ethical pillars were compromised by the school's implementation.
    2. Explain how **Explainable AI (XAI)** and **human final decision-making** could have been used as safeguards to prevent this automated error from harming the student.
    3. Discuss who should hold **accountability** for the mistake.

SECTION C — CONCEPTUAL SHORT-ANSWER QUESTIONS (20 Marks)

Answer the following questions in your own words. Focus on depth of explanation and technical accuracy.

🔬 Question 1: Nanoscale Scaling Limits and Alternative Computing Paradigms (10 Marks)

Explain the physical limitations (specifically **quantum tunneling** and **leakage current**) encountered when silicon circuits are scaled down below nanoscale limits under **Moore's Law**. Then, contrast the fundamental concept of a classical binary **bit** with a quantum **qubit**, explaining how the principle of **superposition** expands the computational state space without requiring further transistor physical miniaturization. [13, 14]

🧠 Question 2: The Core Paradigm Shift in Software Engineering (10 Marks)

Contrast **Conventional Programming** with **Machine Learning**. Rather than simply listing definitions, explain the core conceptual shift in *how* software is built, *what* the programmer's role is in each approach, and *why* conventional programming fails on complex tasks like voice recognition or medical image analysis.

🔑 TEACHER ANSWER KEY & PEDAGOGICAL GUIDE

This section is for teachers, examiners, and self-grading students. It provides verified textbook references, correct answers, and grading rubrics.

SECTION A — MCQ ANSWER KEY

1. Correct Answer: B

- **Explanation:** ENIAC, built using fragile vacuum tubes, represents the "Birth of the Computer" stage in the 1940s–1960s. Apple M1 Ultra and Pentium 4 are modern processors, and Intel 4004 is the first microprocessor (1971).
- **Lesson:** 1-1
- **Textbook Page:** Page 8
- **Objective Tested:** Recall historical stages of IT development.
- **Difficulty:** Easy

2. Correct Answer: C

- **Explanation:** Due to their massive scale, power consumption, and extreme cost, computers during the 1940s-60s were restricted to government, military, and heavy scientific computation. Personal and consumer use did not begin until the 1970s microprocessors.
- **Lesson:** 1-1
- **Textbook Page:** Page 8
- **Objective Tested:** Analyze social impact of early computing.
- **Difficulty:** Medium

3. Correct Answer: B

- **Explanation:** Shrinking the processing units onto a single chip (microprocessor) made computers small and cheap enough to enter office desks and homes as PCs in the 1970s–1980s. Internet, cloud computing, and qubits belong to later stages.
- **Lesson:** 1-1
- **Textbook Page:** Page 8
- **Objective Tested:** Identify key hardware drivers of IT stages.
- **Difficulty:** Easy

4. Correct Answer: C

- **Explanation:** The commercialization of the Internet and the Web, which globalized information and email, occurred in the 1990s.
- **Lesson:** 1-1
- **Textbook Page:** Page 9
- **Objective Tested:** Recall timeline of IT history.
- **Difficulty:** Easy

5. Correct Answer: B

- **Explanation:** The 2000s were defined by the explosive rise of mobile internet and smartphones. ENIAC was the 1940s, PC spread was the 1970s-80s, and cloud computing is the 2010s onward.
- **Lesson:** 1-1
- **Textbook Page:** Page 9
- **Objective Tested:** Identify characteristics of the mobile internet era.
- **Difficulty:** Easy

6. Correct Answer: C

- **Explanation:** Cloud computing is defined as IT delivered over the web as a service, supported by physical datacenters. Edge computing processes data locally; quantum computing uses quantum physics; cashless payments are a social application.
- **Lesson:** 1-1
- **Textbook Page:** Page 10, 23
- **Objective Tested:** Define Cloud Computing.
- **Difficulty:** Easy

7. Correct Answer: C

- **Explanation:** Moore's Law is the empirical observation that the number of transistors on an integrated circuit chip doubles approximately every two years.
- **Lesson:** 1-1
- **Textbook Page:** Pages 9-11
- **Objective Tested:** Recall Moore's Law.
- **Difficulty:** Easy

8. Correct Answer: B

- **Explanation:** Miniaturizing silicon transistors below nanoscale limits causes subatomic problems: quantum tunneling (electrons passing through barriers) and leakage current (heat and current escaping). These represent physical boundaries.
- **Lesson:** 1-1
- **Textbook Page:** Page 13
- **Objective Tested:** Explain why Moore's Law is hitting a wall.
- **Difficulty:** Hard

9. Correct Answer: B

- **Explanation:** Quantum tunneling is a subatomic physical limit where electrons spontaneously slip through extremely thin barriers on a chip, disrupting the logic states and circuit reliability.
- **Lesson:** 1-1
- **Textbook Page:** Page 13
- **Objective Tested:** Define Quantum Tunneling in microprocessors.
- **Difficulty:** Hard

10. Correct Answer: A

- **Explanation:** Leakage current is electrical current that escapes unintentionally from scaled-down circuits, resulting in excessive power consumption and overheating, preventing further performance boosts.
- **Lesson:** 1-1
- **Textbook Page:** Page 13
- **Objective Tested:** Define Leakage Current in silicon scaling.
- **Difficulty:** Hard

11. Correct Answer: D

- **Explanation:** Quantum computing uses the principles of quantum mechanics (superposition) to perform calculations far faster than classical computers.
- **Lesson:** 1-1
- **Textbook Page:** Page 14, multimodal_8
- **Objective Tested:** Recall definitions of emerging technologies.
- **Difficulty:** Easy

12. Correct Answer: B

- **Explanation:** Superposition allows a qubit to represent a combination of 0 and 1 simultaneously. Option A is a classical bit with high voltage, Option C is not how bits work, and Option D refers to conventional software.
- **Lesson:** 1-1
- **Textbook Page:** Page 14
- **Objective Tested:** Contrast classical bits and qubits.
- **Difficulty:** Medium

13. Correct Answer: B

- **Explanation:** AR overlays digital graphics on top of the actual real world (preserving view of the surroundings). VR isolates the user from the real world entirely using an enclosed headset. Option A reverses the definitions.
- **Lesson:** 1-1
- **Textbook Page:** Page 14, multimodal_8
- **Objective Tested:** Distinguish AR from VR.
- **Difficulty:** Medium

14. Correct Answer: A

- **Explanation:** The textbook notes that a delay of even 0.1 seconds in autonomous vehicle communication can lead to a collision, which justifies the necessity of on-board edge computing.
- **Lesson:** 1-1
- **Textbook Page:** Page 14, multimodal_8
- **Objective Tested:** Explain latency thresholds in autonomous driving.
- **Difficulty:** Easy

15. Correct Answer: C

- **Explanation:** Edge computing processes data locally on-board the physical device itself rather than routing it through the cloud.
- **Lesson:** 1-1
- **Textbook Page:** Page 15, multimodal_8
- **Objective Tested:** Define Edge Computing.
- **Difficulty:** Easy

16. Correct Answer: B

- **Explanation:** "The cloud" is not an ethereal concept; it is physical hardware consisting of massive server racks stored in real physical datacenters.
- **Lesson:** 1-1
- **Textbook Page:** Page 10, multimodal_5
- **Objective Tested:** Explain the physical nature of cloud infrastructure.
- **Difficulty:** Medium

17. Correct Answer: B

- **Explanation:** These are the five daily/societal changes directly resulting from the spread of IT (SNS, Cashless Payment, Online Learning, E-Commerce, Remote Work).
- **Lesson:** 1-1
- **Textbook Page:** Pages 10-11
- **Objective Tested:** Contextualize IT social transformations.
- **Difficulty:** Easy

18. Correct Answer: B

- **Explanation:** Social Networking Services (SNS) allow users to connect, post, and share information rapidly. E-commerce is online shopping.
- **Lesson:** 1-1
- **Textbook Page:** Page 11
- **Objective Tested:** Define SNS.
- **Difficulty:** Easy

19. Correct Answer: B

- **Explanation:** A cashless society requires digital tools (cards, phones). Those without access are locked out of buying basic items (social exclusion), highlighting the dark side of IT transformations.
- **Lesson:** 1-1
- **Textbook Page:** Page 23, multimodal_11
- **Objective Tested:** Evaluate social challenges of cashless payment systems.
- **Difficulty:** Medium

20. Correct Answer: B

- **Explanation:** AI is the overarching term for computer technologies reproducing human-like intelligent behavior (reasoning, learning, translating). Option C is conventional programming.
- **Lesson:** 1-2
- **Textbook Page:** Page 31
- **Objective Tested:** Define Artificial Intelligence.
- **Difficulty:** Easy

21. Correct Answer: B

- **Explanation:** AI is the umbrella; Machine Learning is inside AI; Deep Learning is inside ML; Generative AI is inside DL.
- **Lesson:** 1-2
- **Textbook Page:** Page 32, multimodal_16
- **Objective Tested:** Detail the hierarchical relationship of AI technologies.
- **Difficulty:** Medium

22. Correct Answer: C

- **Explanation:** Today's AI is specialized or "narrow" AI—highly expert at one task (e.g., translation) but lacking general human reasoning or intelligence.
- **Lesson:** 1-2
- **Textbook Page:** Page 30, multimodal_15
- **Objective Tested:** Define Narrow AI.
- **Difficulty:** Medium

23. Correct Answer: C

- **Explanation:** Conventional programming relies on humans manually writing rules. Machine learning shifts this paradigm: the computer takes raw data and examples to automatically discover patterns and rules.
- **Lesson:** 1-2
- **Textbook Page:** Page 12, multimodal_19
- **Objective Tested:** Contrast Conventional Programming and Machine Learning.
- **Difficulty:** Hard

24. Correct Answer: D

- **Explanation:** This is the definition of a Neural Network: a system inspired by human brain cells, arranged in layers to learn patterns.
- **Lesson:** 1-2
- **Textbook Page:** Pages 33-34
- **Objective Tested:** Define Neural Networks.
- **Difficulty:** Easy

25. Correct Answer: B

- **Explanation:** ML is a subset of AI that learns rules and statistical patterns directly from data. It does not use hand-written rules (Option D) and is software, not a hardware silicon chip (Option C).
- **Lesson:** 1-2
- **Textbook Page:** Page 33
- **Objective Tested:** Recall Machine Learning role.
- **Difficulty:** Easy

26. Correct Answer: C

- **Explanation:** Generative AI is defined by its ability to *create* brand-new, original content (like images or text) from prompts, rather than just classifying existing data.
- **Lesson:** 1-2
- **Textbook Page:** Page 35, multimodal_18
- **Objective Tested:** Identify examples of Generative AI.
- **Difficulty:** Medium

27. Correct Answer: B

- **Explanation:** Hallucination is when an AI generates highly fluent, plausible-sounding content that is factually incorrect or completely disconnected from reality.
- **Lesson:** 1-2
- **Textbook Page:** Page 38, 54
- **Objective Tested:** Define Hallucination.
- **Difficulty:** Easy

28. Correct Answer: B

- **Explanation:** Generative models do not look up databases of certified facts; they generate content by calculating which words or pixels statistically belong together based on patterns in their training data. This priority on fluency causes hallucinations.
- **Lesson:** 1-2
- **Textbook Page:** Pages 35, 38
- **Objective Tested:** Explain why AI hallucinates.
- **Difficulty:** Hard

29. Correct Answer: B

- **Explanation:** Recommendation systems (like YouTube or Spotify) predict user preferences by analyzing clicking and search history to display personalized content suggestions.
- **Lesson:** 1-3
- **Textbook Page:** Page 49, 51
- **Objective Tested:** Define Recommendation Systems.
- **Difficulty:** Easy

30. Correct Answer: C

- **Explanation:** Voice assistants are on-phone programs (Siri, Google Assistant) that recognize speech coordinates and execute oral instructions.
- **Lesson:** 1-3
- **Textbook Page:** Page 51
- **Objective Tested:** Identify Voice Assistants.
- **Difficulty:** Easy

31. Correct Answer: B

- **Explanation:** Google Translate and DeepL are explicitly named by the textbook as examples of Machine Translation engines.
- **Lesson:** 1-3
- **Textbook Page:** Page 51
- **Objective Tested:** Recall textbook examples of machine translation.
- **Difficulty:** Easy

32. Correct Answer: B

- **Explanation:** Image-diagnosis AI is used in healthcare to analyze scans (CT, X-rays) to assist medical professionals in detecting diseases early.
- **Lesson:** 1-3
- **Textbook Page:** Page 52, 53
- **Objective Tested:** Explain industrial AI in healthcare.
- **Difficulty:** Easy

33. Correct Answer: C

- **Explanation:** Predictive maintenance uses sensors to analyze machine wear and predict failure *before* it happens, saving factory costs.
- **Lesson:** 1-3
- **Textbook Page:** Page 49, 53
- **Objective Tested:** Define Predictive Maintenance.
- **Difficulty:** Easy

34. Correct Answer: B

- **Explanation:** In agriculture, AI is applied to detect crop pests and diseases from photos, and predict harvesting times.
- **Lesson:** 1-3
- **Textbook Page:** Page 53
- **Objective Tested:** Identify AI agricultural applications.
- **Difficulty:** Easy

35. Correct Answer: B

- **Explanation:** The Black-box Problem describes the fact that modern neural networks are so complex that humans cannot trace the exact calculations the model used to reach its final judgment.
- **Lesson:** 1-3
- **Textbook Page:** Page 52, 54
- **Objective Tested:** Define the Black-box Problem.
- **Difficulty:** Medium

36. Correct Answer: B

- **Explanation:** Because a model is an uninspectable "black-box," doctors cannot audit its reasoning to verify if a disease diagnosis is correct or based on biased features, creating serious safety risks.
- **Lesson:** 1-3
- **Textbook Page:** Page 52, 54
- **Objective Tested:** Explain the consequences of the black-box problem.
- **Difficulty:** Hard

37. Correct Answer: C

- **Explanation:** AI excels at pattern matching but lacks an understanding of ethics, responsibility, privacy, and fairness. Thus, human judgment must always verify its outputs.
- **Lesson:** 1-3
- **Textbook Page:** Page 48, 54
- **Objective Tested:** State AI characteristics and boundaries.
- **Difficulty:** Medium

38. Correct Answer: B

- **Explanation:** This is the definition of Algorithmic Bias: systematic errors or prejudices in AI judgments stemming from training data or system design.
- **Lesson:** 1-4
- **Textbook Page:** Page 68, 69
- **Objective Tested:** Define Algorithmic Bias.
- **Difficulty:** Easy

39. Correct Answer: C

- **Explanation:** Algorithmic bias occurs when training data is unbalanced (underrepresenting certain groups) or reflects past human prejudices, which the computer then encodes as patterns.
- **Lesson:** 1-4
- **Textbook Page:** Page 70
- **Objective Tested:** Identify causes of algorithmic bias.
- **Difficulty:** Medium

40. Correct Answer: B

- **Explanation:** A proxy variable is an apparently neutral variable (like a postal code) that correlates so closely with a protected class (like race or income) that the AI reconstructs the bias mathematically.
- **Lesson:** 1-4
- **Textbook Page:** Page 70
- **Objective Tested:** Define Proxy Variables.
- **Difficulty:** Hard

41. Correct Answer: B

- **Explanation:** Even if the gender column is deleted, the AI can find proxy variables in the resumes (such as historical club memberships or women's college names) that stand in for gender, continuing the discrimination.
- **Lesson:** 1-4
- **Textbook Page:** Page 70
- **Objective Tested:** Apply proxy variable concepts to bias.
- **Difficulty:** Hard

42. Correct Answer: B

- **Explanation:** Face recognition is "dual-use" because it offers positive benefits (unlocking phones, catching criminals) but also raises severe democratic threats (unauthorized state tracking and surveillance).
- **Lesson:** 1-4
- **Textbook Page:** Pages 70-71
- **Objective Tested:** Explain dual-use technology.
- **Difficulty:** Medium

43. Correct Answer: B

- **Explanation:** Explainable AI (XAI) consists of tools that translate complex mathematical models into explanations that humans can understand and audit.
- **Lesson:** 1-4
- **Textbook Page:** Page 71, 72
- **Objective Tested:** Define Explainable AI (XAI).
- **Difficulty:** Easy

44. Correct Answer: B

- **Explanation:** XAI provides a technical antidote to the "black-box problem." By making AI decisions transparent, humans can audit them for safety, legality, and fairness.
- **Lesson:** 1-4
- **Textbook Page:** Page 71-73
- **Objective Tested:** Explain the purpose of XAI.
- **Difficulty:** Hard

45. Correct Answer: D

- **Explanation:** Fairness requires that AI systems do not unjustly discriminate against any individual or demographic group.
- **Lesson:** 1-4
- **Textbook Page:** Page 73
- **Objective Tested:** Define the Fairness pillar.
- **Difficulty:** Easy

46. Correct Answer: B

- **Explanation:** Transparency is the ethical requirement to show the AI's decision process, training data, and inner logic clearly to stakeholders.
- **Lesson:** 1-4
- **Textbook Page:** Page 73
- **Objective Tested:** Define the Transparency pillar.
- **Difficulty:** Easy

47. Correct Answer: B

- **Explanation:** Accountability requires establishing clear rules showing who is legally and ethically answerable for the system's outcomes and errors.
- **Lesson:** 1-4
- **Textbook Page:** Page 73
- **Objective Tested:** Define the Accountability pillar.
- **Difficulty:** Medium

48. Correct Answer: C

- **Explanation:** There is currently no globally agreed-upon legal standard for who is liable (developer, business, or operator) when an AI causes harm, which makes accountability a primary debate.
- **Lesson:** 1-4
- **Textbook Page:** Page 72, 86
- **Objective Tested:** Analyze responsibility in AI failures.
- **Difficulty:** Hard

49. Correct Answer: C

- **Explanation:** Facial recognition cameras provide benefits (such as faster automated roll-calls) but threaten personal privacy by tracking students' movements without explicit consent.
- **Lesson:** 1-4
- **Textbook Page:** Page 82, multimodal_39
- **Objective Tested:** Balance convenience and privacy.
- **Difficulty:** Medium

50. Correct Answer: B

- **Explanation:** Privacy Protection requires giving individuals control over how their personal data is collected, stored, and utilized.
- **Lesson:** 1-4
- **Textbook Page:** Page 73, 81
- **Objective Tested:** Define Privacy Protection.
- **Difficulty:** Easy

SECTION B — SCENARIO SOLUTIONS & GRADING RUBRIC

Scenario 1 Model Answer (Cairo Express)

- **Core Task:** Evaluate Edge vs. Cloud Computing for urban collision avoidance in delivery drones using the concept of latency.

Model Reasoning Path:

1. **Analyze latency constraints:** Drones fly at high speeds through narrow urban spaces. To avoid crashing into dynamic obstacles (birds, trees, power lines), decisions must be made instantaneously.
2. **Apply the 0.1-second threshold:** The textbook explicitly states that in autonomous systems (like self-driving vehicles or drones), a delay of even 0.1 seconds can cause a catastrophic accident [14, multimodal_8].
3. **Evaluate Option 1 (Cloud):** Sending video feeds to a remote physical datacenter requires streaming data over a wireless network, waiting for servers to process the data, and waiting for the commands to travel back. Network congestion, signal drops, and physical distance make it impossible to guarantee a latency under 0.1 seconds, creating an unacceptable safety risk.
4. **Evaluate Option 2 (Edge):** Processing data locally on-board the drone bypasses the network entirely. By executing obstacle-detection algorithms on local chips (Edge Computing), the system achieves near-zero latency, allowing immediate collision avoidance under the 0.1-second safety limit.
5. **Recommendation:** The company must choose **Option 2 (Edge Computing)** because local processing is a hard physical requirement to eliminate network latency and ensure drone safety.

Grading Rubric (10 Marks Total):

- **3 Marks:** Correctly selects **Edge Computing (Option 2)**.
- **3 Marks:** Explains the physical nature of **latency** and the **0.1-second safety threshold** in autonomous systems.
- **2 Marks:** Correctly explains why **Cloud Computing (Option 1)** is dangerous (reliance on network transit time, central physical server delays).
- **2 Marks:** Presents a clear, logical, and technically accurate comparison.

Scenario 2 Model Answer (Bank Filter)

- **Core Task:** Explain how an automated recruitment AI discriminated against female applicants even though "Gender" was deleted, using bias and proxy variables.

Model Reasoning Path:

1. **Analyze the training data:** The AI was trained on resumes of employees successfully hired by the bank over the past 20 years. Because historical bank hiring practices may have favored male candidates (historical prejudice), the dataset itself was heavily imbalanced and biased.
2. **Define algorithmic bias:** AI does not have a moral compass; it learns purely by finding mathematical correlations in historical data. The model identified that historical "successful" profiles had specific features that correlated with male resumes.
3. **Explain proxy variables:** Simply deleting the "Gender" column did not make the system neutral. The AI discovered **proxy variables**—neutral data points that correlate heavily with gender.
4. **Identify examples of proxies in resumes:** Examples of proxy variables include:
   - Hobby listings (e.g., "Men's Rowing Team" or specific clubs).
   - Single-gender school names or historical women's colleges.
   - Grammatical patterns or language styles that correlate statistically with past male hires.
5. **Conclusion:** The AI used these proxy variables to mathematically reconstruct the deleted "Gender" attribute. It automated and amplified the bank's historical gender prejudices under the guise of an objective algorithm, demonstrating **algorithmic bias**.

Grading Rubric (10 Marks Total):

- **3 Marks:** Defines **algorithmic bias** and connects it to the bank's **unrepresentative, prejudiced historical training data** (the "Garbage In, Garbage Out" rule).
- **4 Marks:** Defines **proxy variables** and explains how the AI used them to reconstruct the deleted gender attribute (providing realistic examples like school names, sports, or clubs).
- **3 Marks:** Logically explains that deleting a label does not eliminate bias, as complex models easily find patterns in correlated features.

Scenario 3 Model Answer (Attendance Cameras)

- **Core Task:** Analyze the school's gate camera error using the four pillars of AI Ethics, explain safeguards (XAI and human judgment), and discuss accountability.

Model Reasoning Path:

1. **Identify compromised ethical pillars:**
    - **Fairness:** Compromised because an automated error unjustly penalized an innocent student with detention.
    - **Transparency:** Compromised because the school administration hid behind the "black box," stating they did not know how the computer reached its decision and could not explain the error.
    - **Privacy Protection:** Compromised because the system collects, tracks, and processes sensitive biometric data (face scans) of minors at the school gates, apparently without explicit consent or safety audits.
    - **Accountability:** Compromised because the school administration blamed the software and refused to take responsibility for the false absence and detention.
2. **Explain the role of Explainable AI (XAI) as a safeguard:**
    - Instead of an opaque "black box," an **XAI** system would allow school administrators to inspect the decision. It would highlight which features or database conflicts caused the camera to misidentify the student's face coordinates, allowing immediate manual correction.
3. **Explain the role of human final decision-making:**
    - The school made the mistake of leaving the final decision (attendance logging, SMS alert, detention issuance) completely automated. Under the principle of **human final decision-making**, AI should only act as a support tool. A human administrator must verify any flag (such as a missing student or a match error) before issuing disciplinary actions like detention.
4. **Determine Accountability:**
    - The school cannot shift blame to the computer. The school administration chose to deploy the technology and is legally and ethically **accountable** for its outcomes. The developers also share responsibility to test the software for bias and error rates.

Grading Rubric (10 Marks Total):

- **3 Marks:** Correctly identifies and defines at least three compromised ethical pillars (**Fairness, Transparency, Privacy, Accountability**).
- **3 Marks:** Explains how **Explainable AI (XAI)** resolves the "black box" by showing *how* the error occurred.
- **2 Marks:** Explains how **human final decision-making** acts as a safeguard (AI as support, humans verify before issuing detention).
- **2 Marks:** Correctly argues that **accountability** lies with the human institution (the school administration) and developers, not the computer.

SECTION C — SHORT-ANSWER MODEL ANSWERS

Question 1 Model Answer (Nanoscale Scaling Limits and Qubits)

- **Core Task:** Explain silicon scaling limits (quantum tunneling, leakage current) and contrast classical bits with quantum qubits (superposition).

Model Reasoning Path:

- **Nanoscale Silicon Limits:** Moore's Law states that transistor density on integrated circuits doubles approximately every two years [9, 10]. As silicon features approach sub-nanometer levels, they encounter physical barriers that classical scaling cannot overcome:
    1. **Quantum Tunneling:** Logic gate barriers become so thin that electrons spontaneously slip through them, preventing the transistor from reliable switching between on and off states [13].
    2. **Leakage Current:** Unintentional current escapes from the thin logic gates, leading to excessive heat generation and wasted power consumption [13].
- **Classical Bits vs. Qubits:**
    - **Classical Bit:** The basic unit of classical computing. It is a binary switch restricted to holding exactly one state—either **0 or 1**—at any given time [14, multimodal_8].
    - **Qubit:** The basic unit of quantum information. Leveraging quantum mechanics, a qubit can exist in a state of **superposition**, meaning it represents a combination of **both 0 and 1 simultaneously** [14, multimodal_8].
    - This superposition allows quantum systems to maintain an exponentially expanding computational state space, enabling massive parallel operations for specific mathematical problems without depending on physical silicon gate miniaturization.

Grading Rubric (10 Marks Total):

- **2 Marks:** Defines **Moore's Law** and explains that transistor doubling is hitting physical walls.
- **3 Marks:** Explains **quantum tunneling** (electrons slipping through barriers) and **leakage current** (escaping current and heat) as the subatomic causes of this limitation.
- **2 Marks:** Defines **classical bits** as binary units restricted to 0 OR 1.
- **3 Marks:** Defines **qubits** and explains how **superposition** allows them to represent both states simultaneously to achieve massive parallel processing.

Question 2 Model Answer (Programming Paradigm Shift)

- **Core Task:** Contrast Conventional Programming and Machine Learning, detailing the developer's role and why conventional programming fails on complex tasks.

Model Reasoning Path:

- **Conventional Programming:**
  - *The Process:* Humans manually write explicit logic rules (e.g., IF/THEN statements). The computer takes these hand-written rules and **Input Data** to calculate **Output Answers**.
  - *The Developer's Role:* The developer must completely understand the problem and explicitly write out every single rule, mathematical formula, and edge-case scenario.
- **Machine Learning (ML):**
  - *The Process:* We flip the equation. Humans provide the computer with **Input Data** and **Target Answers (Examples)**. The computer's optimization algorithms process these inputs to automatically discover the underlying rules and patterns.
  - *The Developer's Role:* The developer shifts from being a "rule-writer" to a "data-curator." Their role is to select, clean, and balance representative datasets, design the system architecture, and audit the discovered rules for fairness and accuracy.
- **Why Conventional Programming Fails on Complex Tasks:**
  - Complex human tasks—such as recognizing a face in a photo, understanding spoken language with diverse accents, or diagnosing a disease from CT pixels—are **un-rulemable**.
  - It is physically impossible for a human engineer to write explicit mathematical rules to describe every variation in human faces (lighting, angles, facial hair, aging) or natural accents. If a programmer tried to write rules for every scenario, the code would become infinitely complex and immediately break on any unseen variation. Machine learning solves this because its neural networks automatically extract complex visual patterns from raw pixels or waveforms, bypassing the need for manual rules [14, 34].

Grading Rubric (10 Marks Total):

- **3 Marks:** Explains **Conventional Programming** (Input Data + Hand-written Rules $\rightarrow$ Answers) and the developer's role as the manual rule-writer.
- **3 Marks:** Explains **Machine Learning** (Input Data + Target Answers $\rightarrow$ Learned Rules) and the developer's role as a data curator and auditor.
- **4 Marks:** Explains **why conventional programming fails on complex tasks** (human tasks are un-rulemable; humans cannot write explicit code for infinite variations in pixels or waves, whereas ML automatically discovers these patterns from raw data).